

Outlier Detection

Aim

To detect and visualize outliers in a dataset using the **Interquartile Range (IQR) method** in Python, and to understand their effect on data distribution through visual analysis.

Procedure

1. Import the necessary libraries such as pandas, numpy, matplotlib, and seaborn.
2. Load the dataset into a pandas DataFrame for analysis.
3. Perform an initial inspection using `head()`, `info()`, and `describe()` to understand the data and identify numerical features suitable for outlier detection.
4. Define a custom Python function to detect outliers based on the **IQR method**.
5. In the function, calculate the **first quartile (Q1)** and **third quartile (Q3)** for each numerical column.
6. Compute the **IQR** as $Q3 - Q1$ and determine the lower and upper bounds using $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.
7. Identify all data points that fall below the lower bound or above the upper bound as outliers.
8. Apply this function to each selected feature in the dataset to detect the outlier values.
9. Visualize the detected outliers using **boxplots** and **scatter plots** to better understand their distribution.
10. Analyze the visual results to interpret which features contain extreme values and how they affect the overall dataset.

```
import pandas as pd
import numpy as np
arr=np.random.randint(1,100,16)
```

```
arr.mean()
```

```
np.float64(52.625)
```

```
def outlierDet(arr):
    q1,q3=np.percentile(arr,[25,75])
    IQR=q3-q1
    lr=q1-(1.5*IQR)
    ur=q3+(1.5*IQR)
    return lr,ur
```

```
lr,ub=outlierDet(arr)
lr,ub
```

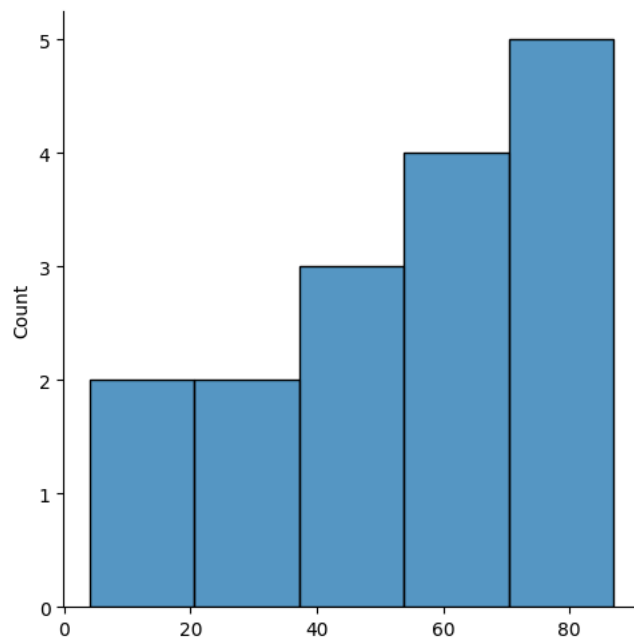
```
(np.float64(-16.25), np.float64(127.75))
```

```
new=arr[(arr>lr)&(arr<ub)]
new
```

```
array([38, 61, 76, 73,  7, 76, 37, 52,  4, 46, 54, 60, 87, 54, 30, 87])
```

```
import seaborn as sns
sns.displot(arr)
```

```
<seaborn.axisgrid.FacetGrid at 0x7826ab3f3dd0>
```



```
sns.distplot(arr)
```

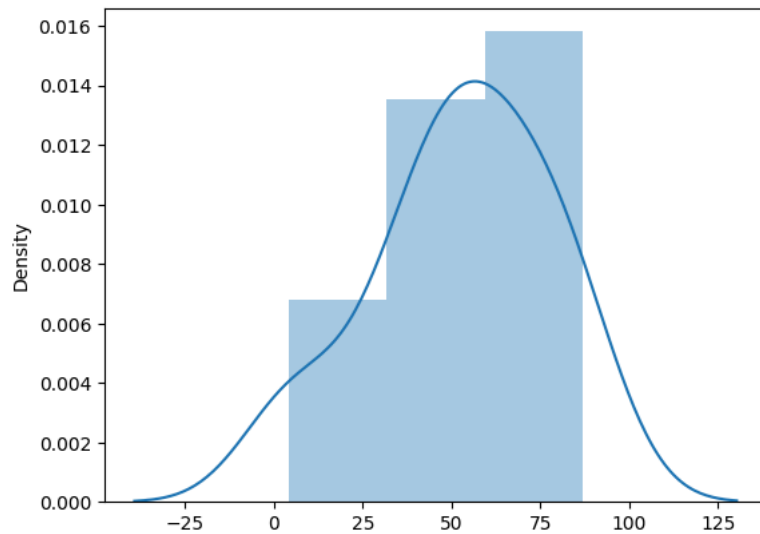
```
/tmp/ipython-input-2700126779.py:1: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(arr)
<Axes: ylabel='Density'>
```



```
sns.displot(new)
```

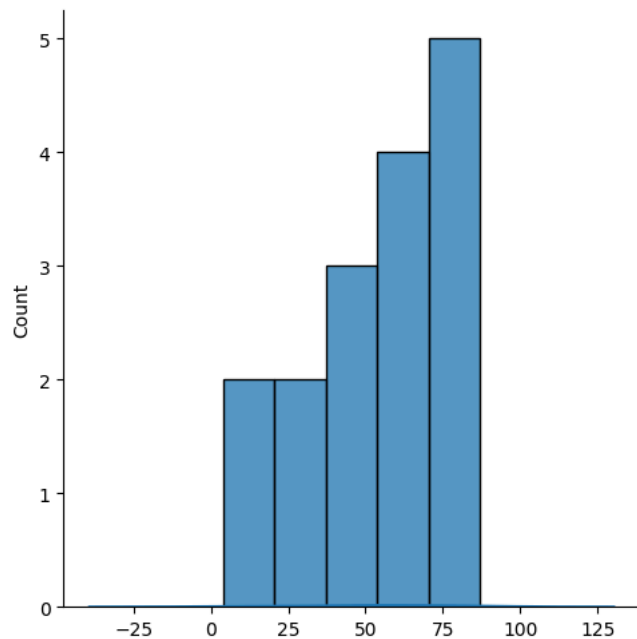
```
/tmp/ipython-input-3821558442.py:2: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(new)
<Axes: ylabel='Count'>
```



```
sns.distplot(new)
```

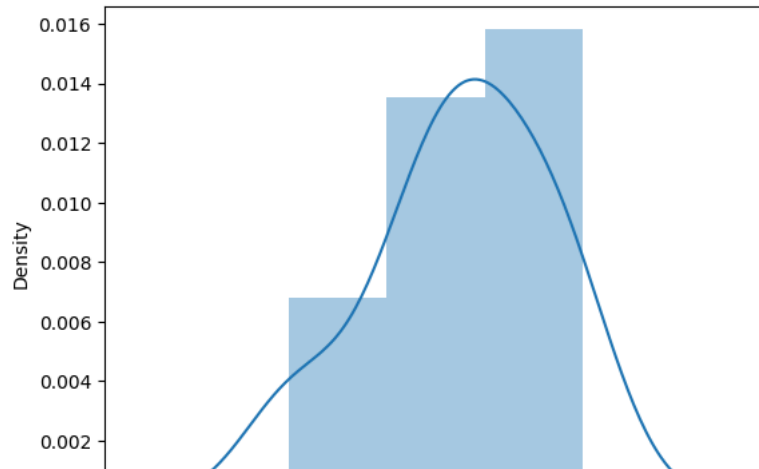
```
/tmp/ipython-input-397519180.py:1: UserWarning:
```

```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(new)  
<Axes: ylabel='Density'>
```



Result:

Hence, a Python program to detect and visualize outliers using the **IQR method** has been successfully implemented, effectively identifying extreme data points for further analysis or cleaning.